All in a Row: Compressed Convolution Networks for Graphs

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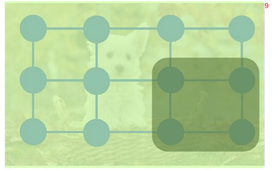
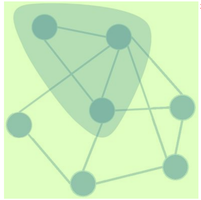
Motivation

Compared to Euclidean convolution, existing graph convolution methods generally fail to learn diverse convolution operators under limited parameter scales and depend on additional treatments of multi-scale feature extraction. In this paper, we want to bridge the gap between Euclidean space and graph space, and therefore ***generalize Euclidean convolution to graphs***.

Generalizing Euclidean convolution to graphs faces challenges from two aspects. From the graph perspective, they generally have **irregular local structures**, different from the grid data in Euclidean space.

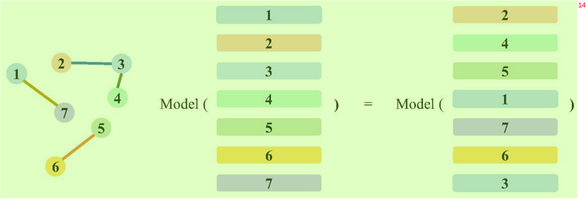


Regular Local Structure



Irregular Local Structure

From the Euclidean convolution perspective, it is **sensitive to the local spatial order**, while GNNs should preserve permutation invariance, i.e., produce the same node representations regardless of the order of the nodes.

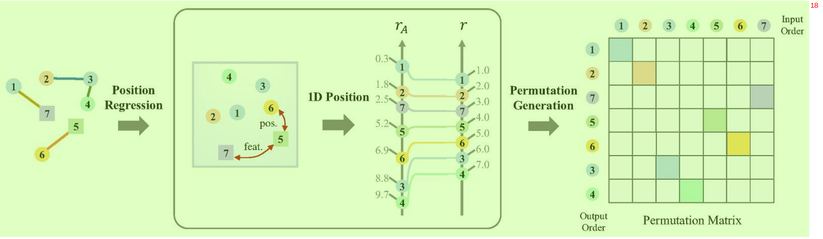


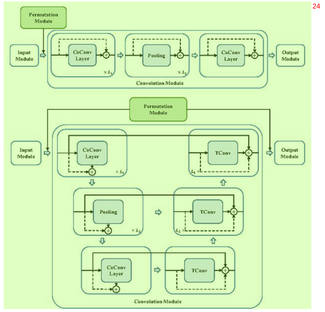
Permutation Invariance / Equivariance

Notations

$\mathbf{X}$	$\mathbf{A}$	$\mathbf{D}$	$\tilde{\mathbf{A}}$	$\mathbf{1}$	n
Node Feat.	Adjacency Mx.	Degree Mx.	$\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}$	All-one Vec.	Node Num.

Method





Regularization on Graphs

We propose to regularize graph-structured data through **learnable permutations** for end-to-end GNNs.

Position Regression: Nodes with similar features or short paths in between will get closer position prediction.

$$\mathbf{r}_A = \tilde{\mathbf{A}}^t \mathrm{MLP}(\mathbf{X}), \quad \mathbf{r} = \mathrm{sgn}\left(\mathbf{r}_A \mathbf{1}^\top - \mathbf{1}\mathbf{r}_A^\top\right)\mathbf{1}. \tag{1}$$

We use the sigmoid function to approximate the sign function in backpropagation.

Permutation Matrix Generation: We resort to the cyclic shift to generate permutation matrices. The differentiable cyclic shift can be implemented through element-wise addition and modulus.

$$\hat{\mathbf{P}} = \exp\left\{-\tau\left[\left(\mathbf{m}\mathbf{1}^\top - \mathbf{1}\mathbf{r}^\top + n\right) \pmod n\right]\right\}, \tag{2}$$

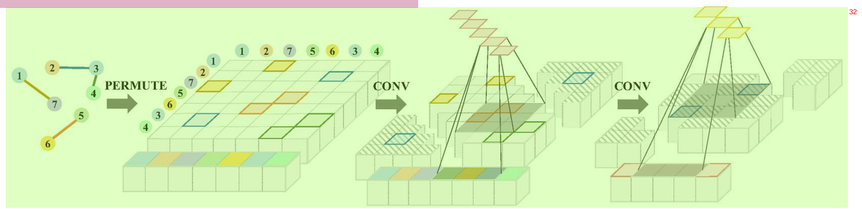
where $\mathbf{m} \in \mathbb{R}^n$ denotes the indicator, initialized as $\mathbf{m}_i = i$, τ denotes the relaxation factor.

Compressed Convolution Network

Learnable permutations **decouple the input order and the follow-up module**, therefore we can apply either permutation-sensitive or permutation-invariant operators on graphs.

The Compressed Convolution Network (CoCN) can be highlighted as:

- Diagonal convolution learns both node features and structure features explicitly.
- Diagonal convolution performs diagonal sliding on structure features to adapt to the scale of graphs.
- CoCNs directly apply hierarchical feature learning on graphs.
- The computational units of CoCNs are nodes/node sets.
- CoCNs use anti-diagonal compression to update structure features.



Main Results

Comparison with State-of-the-Arts

Table 1. **Graph Classification Results (measured by accuracy: %).**

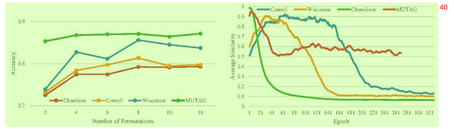
	MUTAG	PROTEINS	NCI1	IMDB-B	IMDB-M	COLLAB
GRAPHS	188	1,113	4,110	1,000	1,500	5,000
AVG NODES	17.93	39.06	29.87	19.77	13	74.5
AVG EDGES	39.6	145.6	64.6	193.1	131.87	4,914.4
NODE FEATURES	7	3	37	0	0	0
PATCHY-SAN	**88.95±4.21**	75.00±2.51	78.60±1.90	71.00±2.29	45.23±2.84	72.60±2.15
GCN	69.50±1.78	73.24±0.73	76.29±1.79	73.26±0.46	50.39±0.41	80.59±0.27
GIN	81.39±1.53	71.46±1.66	80.00±1.40	72.78±0.86	48.13±1.36	78.19±0.63
GRAPHSAGE	83.60±9.60	73.00±4.50	76.00±1.80	68.80±4.50	47.60±3.50	73.90±1.70
PG-GNN	-	76.80±3.80	82.80±1.30	76.80±2.60	53.20±3.60	80.90±0.80
TOPKPOOL	67.61±3.36	70.48±1.01	67.02±2.25	71.58±0.95	48.59±0.72	77.58±0.85
SAGPOOL	73.67±4.28	71.56±1.49	67.45±1.11	72.55±1.28	50.23±0.44	78.03±0.31
ASAP	77.83±1.49	73.92±0.63	71.48±0.42	72.81±0.50	50.78±0.75	78.64±0.50
DIFFPOOL	79.22±1.02	76.25±1.00	62.32±1.90	73.14±0.70	51.31±0.72	82.13±0.43
MINCUTPOOL	79.17±1.64	74.72±0.48	74.25±0.86	72.65±0.75	51.04±0.70	80.87±0.34
STRUCTPOOL	79.50±0.75	75.16±0.86	78.64±1.53	72.06±0.64	50.23±0.53	77.27±0.51
EDGEPOOL	74.17±1.82	75.12±0.76	-	72.46±0.74	50.79±0.59	-
SEP	85.56±1.09	76.42±0.39	79.35±0.33	74.12±0.56	51.53±0.65	81.28±0.15
GMT	83.44±1.33	75.09±0.59	76.35±2.62	73.48±0.76	50.66±0.82	80.74±0.54
COCN (OURS)	87.08±0.17	**76.86±0.13**	**82.89±0.19**	**77.26±0.27**	**56.32±0.18**	**86.15±0.10**

Table 2. **Node Classification Results (measured by accuracy: %).**

	CHAMELEON	SQUIRREL	CORNELL	TEXAS	WISCONSIN	ACTOR	
NODES	2,277	5,201	198	183	251	7,600	
EDGES	36,101	198,493	295	309	499	33544	AVG RANK
FEATURES	2,325	2,089	1,703	1,703	1,703	932	
GCN	59.82±2.58	36.89±1.34	57.03±4.67	59.46±5.25	59.80±6.99	30.26±0.79	11.17
CHEBYNET	55.24±2.76	43.86±1.64	74.30±7.46	77.30±4.07	79.41±4.46	34.11±1.09	8.17
GEOM-GCN	60.00±2.81	38.15±0.92	60.54±3.67	66.76±2.72	64.51±3.66	31.59±1.15	10.17
GRAPHSAGE	49.06±1.88	36.73±1.21	80.08±2.96	82.03±2.77	81.36±3.91	35.07±0.15	7.00
FAGCN	46.07±2.11	30.83±0.69	76.76±5.87	76.49±2.87	79.61±1.58	34.82±1.35	9.17
APPNP	40.44±2.02	29.20±1.45	56.76±4.58	55.10±6.23	54.59±6.13	30.02±0.89	13.00
MIXHOP	60.50±2.53	43.80±1.48	73.51±6.34	77.84±7.73	75.88±4.90	32.22±2.34	8.33
LINKX	68.42±1.38	61.81±1.80	77.84±5.81	74.60±8.37	75.49±5.72	36.10±1.55	6.33
GGCN	71.14±1.84	55.17±1.58	85.68±6.63	84.86±4.55	86.86±3.29	**37.54±1.56**	2.50
ACMII-GCN	68.46±1.70	51.80±1.50	85.95±5.64	**86.76±4.75**	**87.45±3.74**	36.43±1.20	2.33
GPRGCN	62.59±2.04	46.31±2.46	78.11±6.55	81.35±5.32	82.55±6.23	35.16±0.90	5.17
GCNII	63.86±3.04	38.47±1.58	77.86±3.79	80.39±3.83	77.57±3.40	37.44±1.30	5.83
COCN (OURS)	**79.17±0.17**	**72.95±0.23**	**86.22±0.49**	85.21±0.49	86.88±0.45	36.35±0.12	**1.83**

Partial Model Study Results and Key Conclusions

Ablation studies on the number of permutations (left), and output similarity among different permutation heads during training (right).



Takeaways:

- Permutation module can permute input nodes with little information loss. The larger the relaxation factor τ becomes, the less the information loss is.
- CoCNs learn complementary feature representations under different permutations and require multiple permutations (around 8-16) for sufficient feature learning.
- CoCNs achieve better performance as the receptive field size increases (bounded by the number of input nodes).
- Both node features and structure features generally contribute to the model prediction.

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